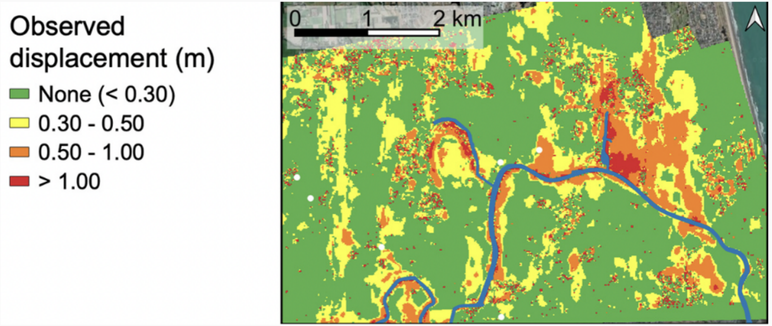
Exploratory Data Analysis

Krishna Kumar & Ellen Rathje

University of Texas at Austin

Overview

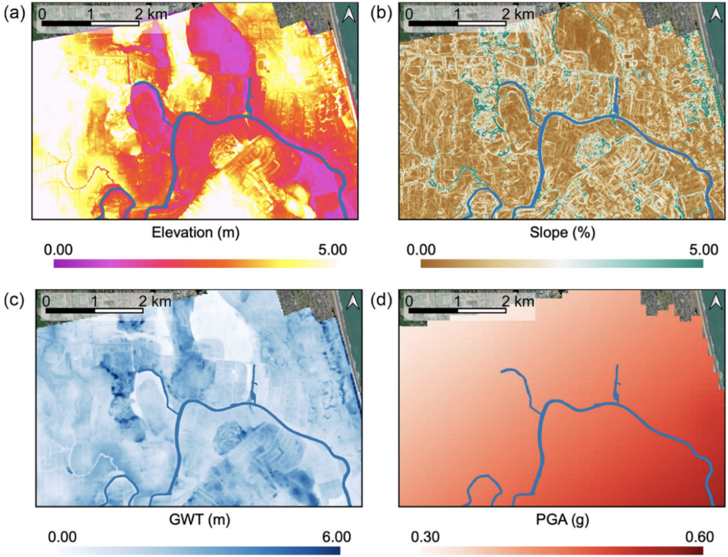
Predicting Lateral Spreading in New Zealand



Goal: Predict lateral spreading displacement during earthquakes

Reference: Durante, M. G., & Rathje, E. M. (2021). An exploration of the use of machine learning to predict lateral spreading. *Earthquake Spectra*, 37(4), 2288-2314.

Input Features for Prediction



The Foundation of Machine Learning

Before building models, understand your data!
EDA answers:

- What does the data tell us about the physics?
- Are there quality issues (missing, errors, outliers)?
- Which features predict lateral spreading?
- How should we preprocess?

The Key Question

Dataset: Durante & Rathje (2021)

- 7,291 case histories
- Target: $> 0.3m$ displacement $=$ spreading

Can we distinguish spreading from non-spreading sites?

Approach: Compare distributions by class

- **Overlapping distributions:** Feature not useful
- **Separated distributions:** Feature is predictive

Best visualization: Overlaid histograms

- See both classes simultaneously
- Directly compare shapes and locations
- Immediate intuition about separability

This is more informative than looking at all data together!

Reading Overlaid Histograms

What to look for:

1. **Separation:** How far apart are the peaks?
   - Far apart → strong predictor
   - Overlapping → weak predictor
2. **Direction:** Which class has higher values?
   - GWD: No-spread sites have higher values (deeper water table = safer)
   - PGA: Spread sites have higher values (more shaking = more risk)
3. **Overlap area:** How much do they overlap?
   - Small overlap → easy to classify
   - Large overlap → difficult boundary

Why Quartiles Matter

Problem: Mean and std are sensitive to outliers
Example: Water depths (m): $\{1.5, 1.8, 2.0, 2.1, 2.3, 15.0\}$

- Mean $= 4.12$ m (misleading!)
- Median $= 2.05$ m (representative!)

Quartiles divide data into four equal parts:

- Q1 (25th percentile): 25% below
- Q2 (Median, 50th percentile): middle value
- Q3 (75th percentile): 75% below
- IQR = Q3 - Q1 (middle 50%)

Box plots visualize the five-number summary

Quartiles for Engineering Decisions: What do these numbers mean?

Example: Ground Water Depth (GWD)
From our data:

- Q1 = 1.64 m, Median = 1.98 m, Q3 = 2.46 m
- IQR = 0.82 m

Engineering interpretation:

- 50% of sites: water table at 1.64-2.46 m depth
- Sites with GWD $<$ Q1 (1.64 m): **High risk** zone
- Sites with GWD $>$ Q3 (2.46 m): **Lower risk** zone

Action: Quartiles create natural risk categories!

The Outlier Dilemma

In geotechnical engineering, outliers can be:

1. **Measurement errors:** Remove
   - Negative water table depth
   - Physically impossible values
2. **Extreme events:** Keep!
   - Major earthquakes
   - Unusual but valid site conditions

Never blindly remove outliers!

- They might be your most important data
- Use engineering judgment + statistics

IQR Method (Most Robust)

Rule: Values beyond these bounds are outliers

$$\text{Lower bound} = Q1 - 1.5 \times \text{IQR}$$
$$\text{Upper bound} = Q3 + 1.5 \times \text{IQR}$$

Why 1.5×IQR?

- John Tukey's empirical rule
- For normal distribution: captures ~99.3% of data
- Balances sensitivity vs over-flagging

What's a good outlier percentage?

- 1-5%: Excellent - typical for clean data
- 5-10%: Good - acceptable natural variability
- >10%: Investigate - may indicate data quality issues

IQR Results for Our Dataset

Outlier analysis using IQR method:

Feature	IQR	Outliers	%	Assessment
GWD	0.82 m	133	1.8%	Excellent
L	0.99 km	8	0.1%	Excellent
Slope	0.95%	542	7.4%	Good
PGA	0.06 g	3	0.0%	Excellent

Analysis:

1. **GWD (1.8%):** Valid extreme water table depths → Keep
2. **L (0.1%):** Very few unusual distances → Keep
3. **Slope (7.4%):** Natural field variability → Keep
4. **PGA (0.0%):** Perfect - earthquake data well-controlled → Keep

Decision: All outliers $< 10\%$ and physically plausible → **Keep all data**
Why? Model needs to learn extreme events (major earthquakes, steep slopes)

What is a Z-Score?

Z-score = how many standard deviations from the mean

$$z = \frac{x - \mu}{\sigma}$$

Interpretation:

- $z = 0$: At the mean
- $z = 1$: One std above mean
- $z = -2$: Two std below mean
- $|z| > 3$: Outlier (only 0.3% of normal data)

68-95-99.7 Rule:

- 68% within $\pm 1\sigma$
- 95% within $\pm 2\sigma$
- 99.7% within $\pm 3\sigma$

IQR vs Z-Score: When to Use Each

Method	Use For	Advantage
IQR	Outlier detection	Robust
Z-score	Outlier detection	Assumes normality
Z-score	**Standardization**	**ML preprocessing**

Recommendation:

- Detect outliers with IQR
- Standardize features with z-scores for neural networks
- Tree-based models don't need standardization

Correlation Coefficient

Measures linear relationship strength
Properties:

- Range: $r \in [-1, 1]$
- $r = 1$: Perfect positive correlation
- $r = -1$: Perfect negative correlation
- $r = 0$: No linear correlation

Interpretation for liquefaction:

- GWD vs Target: Negative (shallow water = more risk)
- PGA vs Target: Positive (more shaking = more risk)
- L vs Target: Negative (closer to river = more risk)

Warning: Correlation $\neq$ Causation!

Mann-Whitney U Test

Question: Are distributions truly different, or just random chance?
What is U? U counts the "wins" - how often one group beats the other.

The Algorithm:

1. **Mix all values** from both groups together
2. **Rank them** from smallest to largest (1, 2, 3, ...)
3. **Sum ranks** for each group
4. **Calculate U:** How often does spreading value ¡ non-spreading value?

Why it's powerful:

- No assumptions about distribution shape
- Robust to outliers (uses ranks, not values)
- Perfect for comparing two groups

Mann-Whitney U Test - Visual Example

Example with GWD (Ground Water Depth):

```
All sites mixed and ranked by GWD:
Spreading sites:      [1.2] [1.5] [1.8] ...
                        ↓     ↓     ↓
                     rank:1 rank:2 rank:3
Non-spreading sites:  [2.5] [3.0] [3.2] ...
                        ↓     ↓     ↓
                     rank:4 rank:5 rank:6
→ Spreading sites get LOWER ranks
→ They have shallower water tables!
```

The p-value answers:

"If groups were actually the same, what's the chance we'd see this much separation by random luck?"

- $p < 0.001 \rightarrow$ Less than 0.1% chance $\rightarrow$ **Strong evidence!**
- $p > 0.05 \rightarrow$ More than 5% chance $\rightarrow$ Could just be randomness

Mann-Whitney Results for Our Data

Statistical significance of each feature:

Feature	p-value	Significant?	Conclusion
GWD	4.76e-15	***	Strong predictor
L	4.90e-46	***	**Strongest** predictor
Slope	9.83e-02	No	Weak predictor
PGA	1.35e-16	***	Strong predictor

Key Insight:

- **Slope is NOT significantly different** $\rightarrow$ May not help prediction!
- **GWD, L, PGA** show extremely low p-values $\rightarrow$ Key predictors

P-value interpretation:

- $p < 0.001$: *** Extremely significant
- $p < 0.01$: ** Very significant
- $p < 0.05$: * Significant
- $p \geq 0.05$: Not significant

Quick Quality Checks

Before modeling, verify:

1. **Missing values:**
    - How many? Which features?
    - Delete ($< 5\%$) or impute ($> 5\%$)?
2. **Physical plausibility:**
    - Negative values where impossible
    - Values outside engineering bounds
3. **Duplicates:**
    - Exact copies (data entry errors)?
    - Same site, multiple events (valid)?
4. **Class balance:**
    - 50-50 split? 90-10?
    - Severe imbalance needs special handling

Common Mistakes to Avoid

1. **Blindly trusting summary statistics**
   - Always visualize!
2. **Removing outliers without investigation**
   - Might be your most important data
3. **Confusing correlation with causation**
   - Ice cream sales correlate with drowning... both caused by summer!
4. **Ignoring class imbalance**
   - 99% accuracy meaningless for 1% rare class
5. **Skipping domain knowledge check**
   - Does this make engineering sense?

Feature Engineering

Creating new features from existing data
What features to introduce for liquefaction datasets

1. **Interaction features:**
   - GWD $\times$ PGA (water depth $\times$ shaking intensity)
   - L $\times$ Slope (distance $\times$ slope)
2. **Polynomial features:**
   - GWD^2, PGA^2 (capture non-linear relationships)
3. **Domain-specific features:**
   - L/GWD (normalized distance)
   - is_shallow_water = GWD < 2 m (binary threshold)
4. **Binning/Discretization:**
   - Convert GWD: "shallow", "medium", "deep"
5. **Transformations:**
   - $\log(L)$, $\sqrt{\text{Slope}}$ (handle skewness)

Summary

EDA is the foundation of successful ML

We accomplished:

- Understood data structure and quality
- Identified predictive features
- Compared distributions by class
- Detected and evaluated outliers
- Gained physical insight into liquefaction
- Made informed preprocessing decisions

"No amount of sophisticated modeling can compensate for poor understanding of your data."

Questions?

"All models are wrong, but some are useful."

— George Box

"See the structure first, then calculate."

— Gilbert Strang

Contact:
Dr. Krishna Kumar [krishnak@utexas.edu]
Dr. Ellen Rathje [rathje@mail.utexas.edu]
University of Texas at Austin

Remember:
"Understanding the data is understanding the physics."